

HEET MEHTA

New York, NY

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LinkedIn

GitHub

Portfolio

EDUCATION

New York University, Tandon School of Engineering

Sep 2025 to May 2027

Master of Science in Computer Science, **GPA: 3.83/4.0**

Brooklyn, NY

Courses: Design & Analysis of Algorithms, Software Engineering, Machine Learning, Big Data

Vellore Institute of Technology

Sep 2021 to Jun 2025

Bachelor of Technology in Information Technology, **GPA: 9.14/10**

Tamil Nadu, India

Courses: Data Structures and Algorithms, Linear Algebra, Probability and Statistics, Numerical Methods,

Operating Systems, Computer Networks, Artificial Intelligence, Cloud Computing

SKILLS

Languages : Python, C++, Java, Go, SQL, Bash, TypeScript

Algorithms and Systems : Algorithm Design, Distributed Systems, Real-Time Processing, Concurrency, Performance Optimization

Numerical and Analytical : NumPy, SciPy, Pandas, Statistical Modeling, Kalman Filtering, Signal Processing, PyTorch

Infrastructure : Linux, AWS, GCP, Kubernetes, Docker, Bazel, Kafka, Redis, ClickHouse, CI/CD

Practices : Clean and Efficient Code, Attention to Detail, Problem Solving, Code Reviews, Technical Documentation

EXPERIENCE

Intrinsic (AI Robotics, Google) | Software Engineering Intern

May 2025 to Aug 2025

- Designed and built a real-time benchmarking system in Go to evaluate scalable backend infrastructure for a production robot fleet, systematically comparing Prometheus, VictoriaMetrics, Thanos, and Mimir under loads exceeding 782k time series with precise measurement of ingestion rate, memory behavior, and latency.
- Engineered a containerized load generator via Bazel deployed on Kubernetes with Helm, producing reproducible results across all configurations; applied strong attention to detail to ensure benchmark methodology was defensible and auditable by senior engineers.
- Delivered a structured technical report synthesizing quantitative findings that directly informed an infrastructure migration decision; worked independently while collaborating effectively with quants and senior developers throughout.

Bhabha Atomic Research Centre | Project Intern

Dec 2024 to May 2025

- Applied numerical computing techniques in Python to fuse noisy sensor data from 500+ IoT devices using RSSI-based trilateration and Kalman filtering, achieving 1-meter indoor positioning accuracy under real-world interference conditions.
- Built multithreaded real-time data pipelines and automated simulation scripts to validate system behavior at scale, reducing evaluation time by 30% while maintaining correctness of statistical estimates.

Encardio Rite | Software Development Intern

May 2024 to Aug 2024

- Built an ML-powered diagnostics system in Python using Scikit-learn on AWS Lambda, applying statistical anomaly detection across sensor streams from 20+ installations and reducing critical failures by 40%.
- Designed scalable data ingestion pipelines integrating multi-source sensor data via MQTT and AWS IoT Core; applied signal processing and feature engineering improving fault detection accuracy by 20%.

PROJECTS

Aegis: Distributed Job Queue & DAG Orchestrator | Go, TypeScript, React, PostgreSQL, Docker

Link

2025

- Designed and built a self-hosted distributed job queue and DAG workflow orchestration engine in Go, handling job leasing with automatic timeout-based reassignment for crashed workers, retries with exponential backoff and jitter, and a dead-letter queue for permanently failed jobs.
- Built a DAG-based workflow engine with cycle detection and fan-out/fan-in advancement; shipped a live React dashboard, Go/Python worker SDKs, and Postgres-based leader election for multi-node high availability.

Eco-Nexus: Multi-Agent Decision System | Python, Next.js, TypeScript, Node.js, Snowflake

Link

2025

- Built an agentic AI engine with explainable multi-variable scoring across 30+ inputs improving decision accuracy by 25%; designed for real-time state sync via Socket.io supporting 50+ concurrent sessions. Awarded 3rd Place at HackNYU 2026.

PUBLICATIONS

- Published “Applications of Machine Learning in Detecting Unethical Sources of Raw Materials in Supply Chains in the Cosmetic Industry” at ICSES 2024, IEEE Xplore. Link

LEADERSHIP & SOCIAL IMPACT

- Volunteer Web Developer, Catchafire Link — Led pro bono website redesign supporting workforce reintegration programs, improving navigation and accessibility for community impact.
- Technical Team Lead, VIT Robotics Club — Mentored 8 members across software and analytical projects, organized 5 technical workshops on algorithms and engineering fundamentals.